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Assignment 1: Data Preprocessing Report

1. Choose Domain Area:

- 1.1. The data set that we have chosen belongs to the domain of healthcare. It deals with patients suffering from lung disease and is the patient recovered after the treatment after the hospital visit. It includes the following 8 data attributes:

Attribute	Type	Description
Age	float64	disease between different age groups.
Gender	object	disease across different genders.
Smoking Status	object	patient being a smoker or not.
Lung Capacity	float64	used to access disease severity.
Disease Type	object	categorized into either COPD or Bronchitis.
Treatment Type	object	categorized into therapy, medication or surgery.
Hospital Visits	float64	number of visits to the hospital for managing the condition.
Recovered	object	did the patient recover after the treatment or not?

2. Documenting Summary as a Report:

- 2.1. Techniques for preprocessing:

2.1.1. Loading the CSV file into the program:

- 2.1.1.1. We begin by creating a data frame using the *pandas* library using our dataset (lung_disease_data.csv) through the built in *read_csv()* function. It then prints out the columns, number of rows and columns and then the first 5 rows of the data frame.

```
.....
.....
Making Tables From CSV
.....
.....
Index(['Age', 'Gender', 'Smoking Status', 'Lung Capacity', 'Disease Type',
      'Treatment Type', 'Hospital Visits', 'Recovered'],
      dtype='object')
We have 5200 rows
We have 8 columns

   Age  Gender Smoking Status  Lung Capacity Disease Type Treatment Type  Hospital Visits  Recovered
0  71.0  Female           No         4.49      COPD      Therapy          14.0         Yes
1  34.0  Female           Yes         NaN      Bronchitis  Surgery           7.0         No
2  80.0   Male           Yes         1.95      COPD      NaN           4.0         Yes
3  40.0  Female           Yes         NaN      Bronchitis  Medication          1.0         No
4  43.0   Male           Yes         4.60      COPD      Surgery           NaN         Yes
```

- 2.1.1.2. We then count the number of missing values in each column. We have an array to hold the columns with missing values and we loop through the data frame column by column seeing if it contains any missing values. We have a

running *total* which holds the total number of missing values in that column which we print out and if the total is greater than 0, we add the name of that column to our array. We repeat these steps for the rest of the columns. Finally, we print out the total number of missing values in each column along with which columns had the missing data.

```
.....
Missing Values In Each Column
.....
Age: 300
Gender: 300
Smoking Status: 300
Lung Capacity: 300
Disease Type: 300
Treatment Type: 300
Hospital Visits: 300
Recovered: 300

These columns had missing data: ['Age', 'Gender', 'Smoking Status', 'Lung Capacity', 'Disease Type', 'Treatment Type', 'Hospital Visits', 'Recovered']
```

- 2.1.1.3. After counting the missing values, we drop all the rows with no data using the built in *dropna()* function. We decided this way was easier than populating the rows with the median data for the numerical columns (Age, Lung Capacity, Hospital Visits) or mode for the categorical ones (Gender, Smoking Status, Disease Type, Treatment Type, Recovered).

```
.....
Dropping Rows With Missing Data
.....
We have 5200 [original] rows
We have 8 [original] columns

   Age  Gender Smoking Status  Lung Capacity Disease Type Treatment Type Hospital Visits Recovered
0  71.0  Female             No           4.49         COPD         Therapy           14.0         Yes
1  34.0  Female             Yes           NaN      Bronchitis         Surgery           7.0         No
2  80.0   Male             Yes           1.95         COPD             NaN           4.0         Yes
3  40.0  Female             Yes           NaN      Bronchitis         Medication           1.0         No
4  43.0   Male             Yes           4.60         COPD         Surgery           NaN         Yes

We have 3236 [original] rows
We have 8 [original] columns

   Age  Gender Smoking Status  Lung Capacity Disease Type Treatment Type Hospital Visits Recovered
0  71.0  Female             No           4.49         COPD         Therapy           14.0         Yes
5  22.0  Female             No           3.65      Bronchitis         Medication           11.0         Yes
6  41.0   Male             Yes           2.98         Asthma         Medication           12.0         Yes
8  21.0  Female             No           4.05  Lung Cancer         Medication           12.0         No
9  49.0   Male             Yes           5.70      Pneumonia         Medication           11.0         Yes
```

- 2.1.1.4. Next part was to remove any duplicate data as it can skew the distribution of the dataset. We attained this by using panda's built in *drop_duplicates()* function with the parameters "keep = false". This option removes all the rows which have duplicates in them. We chose this option to truly have a unique dataset as it

better serves to remove the noise from the dataset.

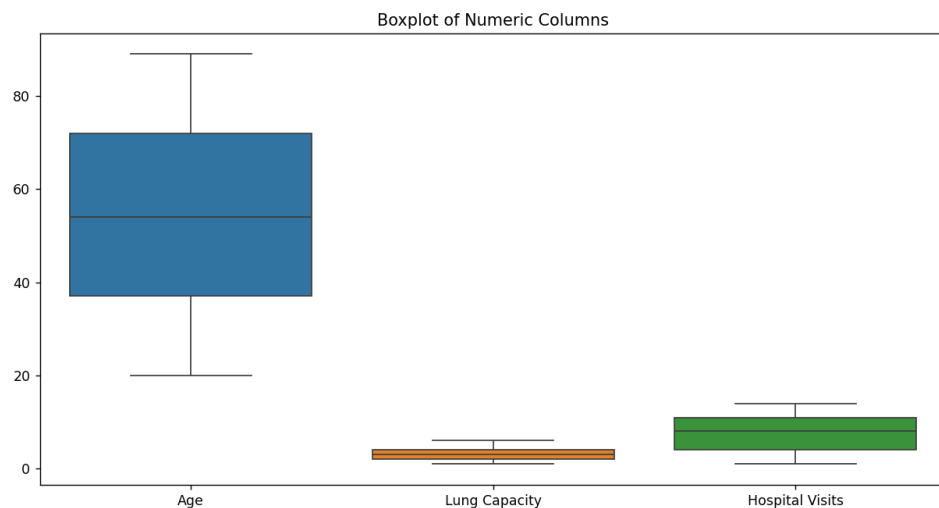
```
.....
.....
Removing Duplicates
.....
.....

Numer of rows WITH duplicates: 3236

Numer of rows WITHOUT duplicates: 3056

There were 180 duplicates in the data frame
```

2.1.1.5. We did not need to remove any outliers because our boxplot did not show any data points below the 25% zone or above the 75% zone.



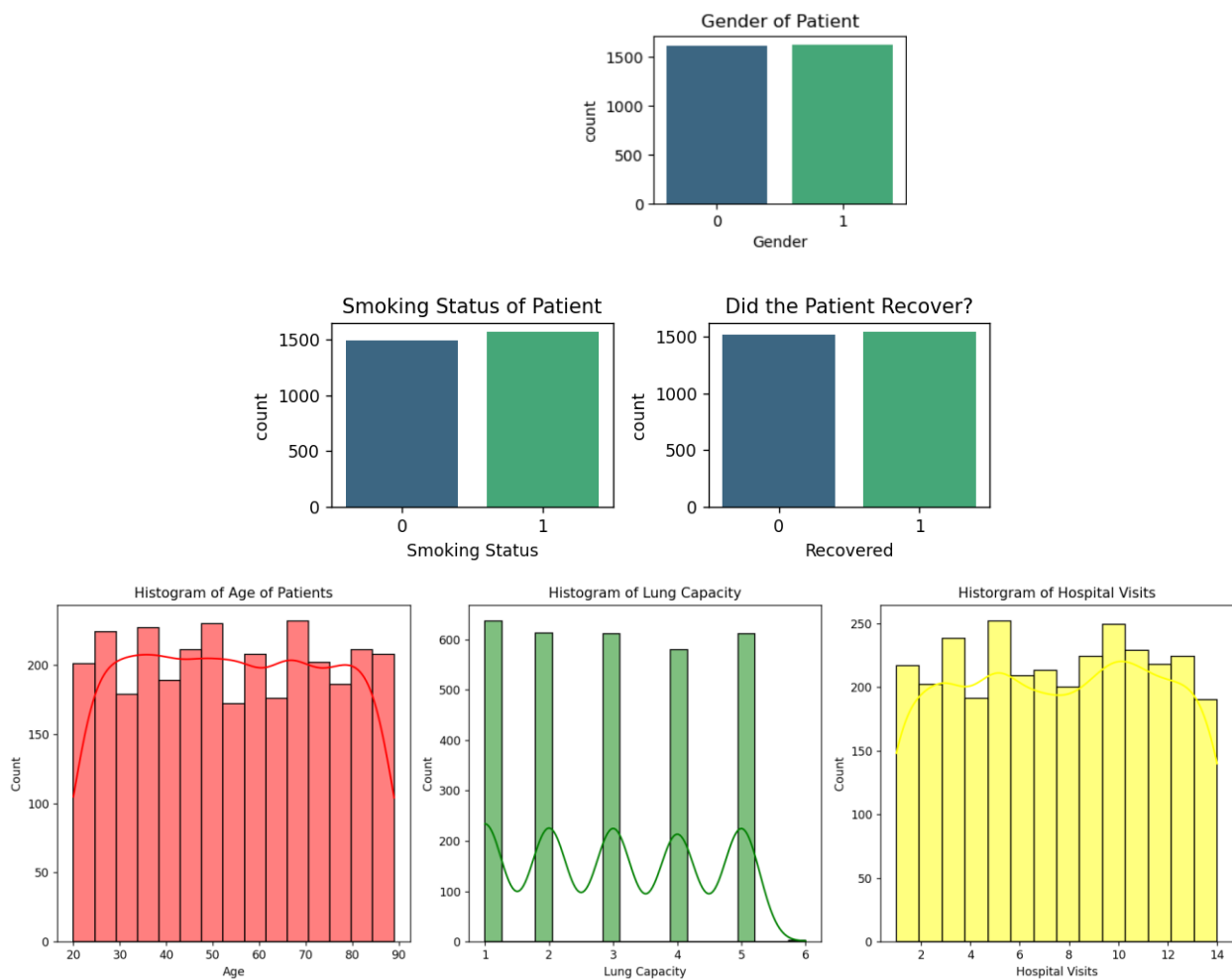
2.1.1.6. Two of the columns (Disease and Treatment Types) had categorical data which needed further breakdown as they contained more than 2 attributes. Hence, we used one hot encoding which was able to break down the two columns into their respective sub-columns. For example, the Disease Type column got broken into; Disease Type_Bronchitis, Disease Type_COPD, Disease Type_Lung Cancer and Disease Type_Pneumonia. Hot encoding enabled us to have the value of 1, for the disease which the patient had while making the rest of columns 0, for that row. The same technique was used for the Treatment column. We then saved this dataset into a new CSV file called *lung_disease_cleaned.csv*.

```

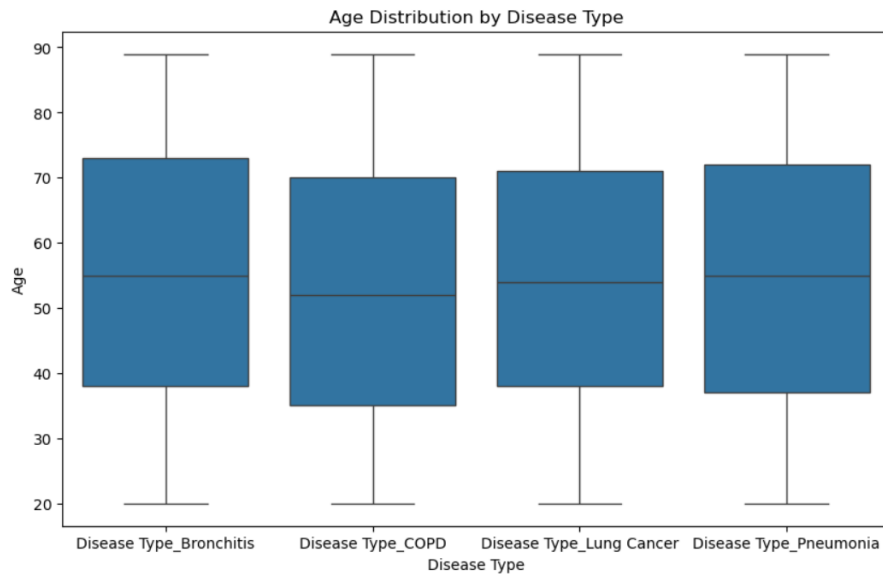
.....
      Converting Categorical Columns to Numeric
.....
.....
Gender Column.....DONE
Smoking Status Column.....DONE
Recovered Column.....DONE
Disease Column.....DONE
Treatment Column.....DONE
>>> Data saved to lung_disease_data_numeric.csv

```

2.1.1.7. We then plotted the numerical data all by themselves to visualize any distinction between the attributes.



2.1.1.8. We then plotted the different columns against the other columns to see the relationship between them and to provide our interpretation of the data as well as our insights.



Interpretation

It seems that patients suffering from Bronchitis had 1/4 of them of ages between 20 - 38 and the other 1/4 between 74 - 89. The median age group (1/2) suffering from this disease were about 55.

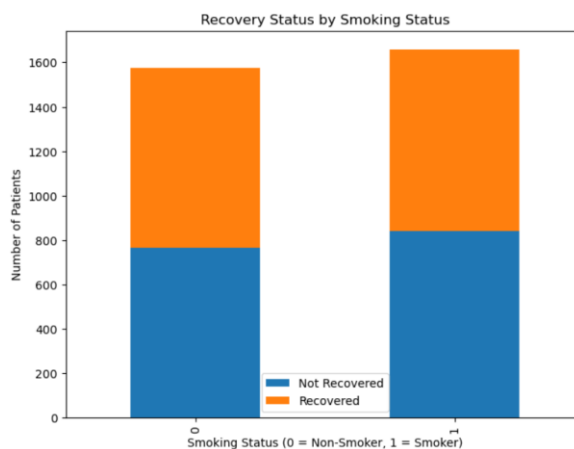
For COPD, the first 1/4 were between 20 -35 and their upper 1/4 were between 70 - 89 while the median age was about 53.

For Lung cancer, the first 1/4 were between 20 -35 and their upper 1/4 were between 70 - 89 while the median age was about 54.

For Pneumonia, the first 1/4 was between 20 -38 and their upper 1/4 were between 72 - 89 while the median age was about 55.

Insights

Based on the interpretation above, we can predict that people who suffer from lung diseases majority are in their mid-50's with the older age group (greater than 70) suffering from Bronchitis and Pneumonia.



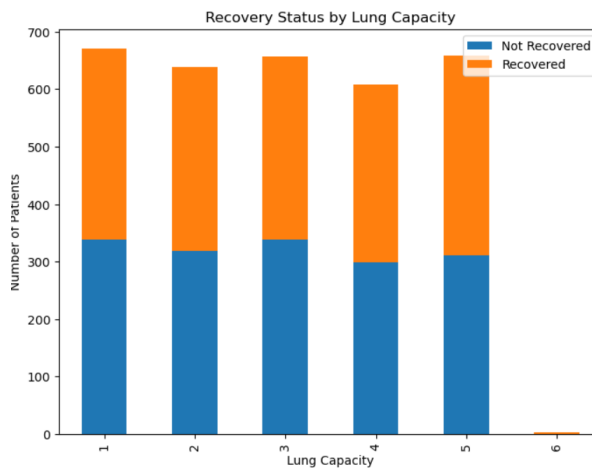
Interpretation

The stacked bar graph depicts that patients who were Non-smokers had about similar recovery outcomes as the ones who smoked.

However, we can clearly see that Smokers had a higher outcome of not recovering from their lung diseases.

Insight

The insight that we can gain from this is that smoking can worsen the recovery outcome of lung diseases.



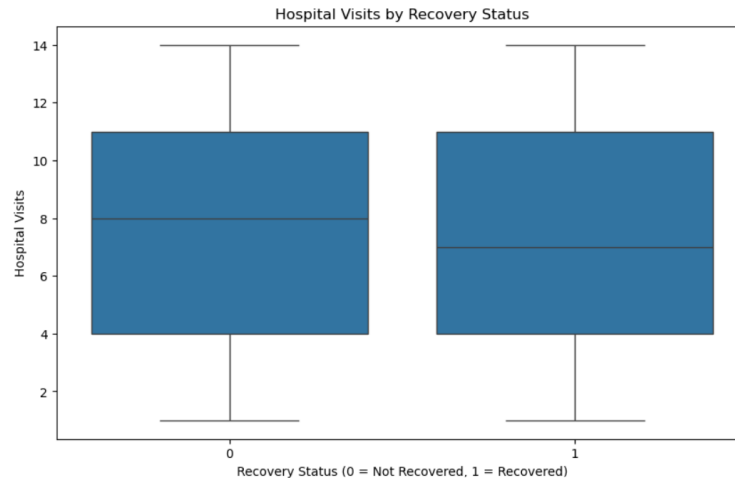
Interpretation

The stacked bar graph shows that patients who had a lower lung capacity (1, 2 and 3) had higher "Not recovered" count.

Moreover, patients with a high lung capacity (4-5) had about similar ratio of Recovered and Not Recovered. Finally, patients with the highest lung capacity of 6 had always recovered.

Insight

The insight that we can make from this is that a lower lung capacity can significantly reduce the recovery outcomes whereas a very one can almost always guarantee recovery.



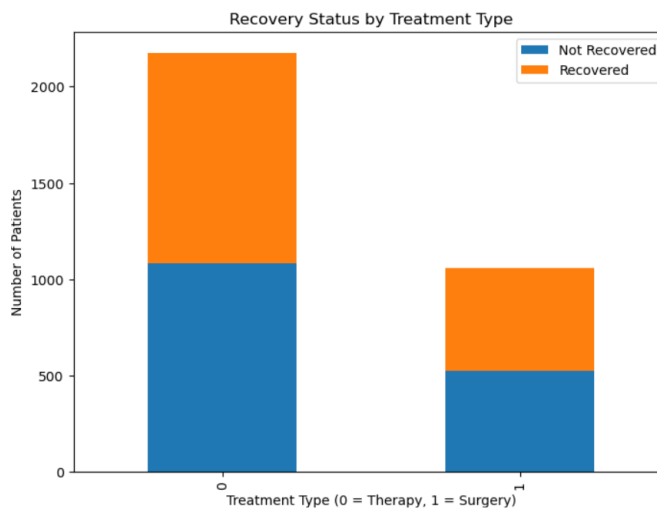
Interpretation

We can see that 1/4 of the patients who had not recovered had hospital visits between 2-4 and the upper 1/4 we between 11-14 visits. Most of the patients had about 8 visits.

In contrast, patients who did recover had a median visit of about 7.

Insight

We can infer from the interpretation above that patients who have a hard time treating their conditions would require more hospital visits.



Interpretation

We can see that patients which opted for therapy as their form of treatment had a higher count of not recovering when compared to the ones who chose surgery. The ones who chose surgery had a higher number of recovered after the treatment.

Insight

We can say that surgical forms of treatment work much better than just therapeutical kind.

3. Developing and documenting human insights with human interpretation on preprocessed data and possible effect on predictions:

Looking at the data, smokers had a higher risk of developing COPD lung disease, and this warranted more hospital visits. Furthermore, patients with lower lung capacity got associated with more lung diseases and they were shown to have a lower recovery rate. Moreover, patients with whomever chose the surgical method of treatment were shown to have more severe disease conditions. In contrast, the ones who choose therapy may be afflicted with the acute stage of the disease. Finally, the patients who had an older age were shown to be suffering from chronic diseases like COPD or lung cancer whereas younger ones had acute conditions such as Pneumonia.

How would this affect our predictions? It could show that age and smoking status might be the two strong factors warranting disease severity (acute or chronic). The lung capacity could serve as a primary factor for the recovery of a patient as well as the treatment type with chronic cases needing treatment options such as surgeries.

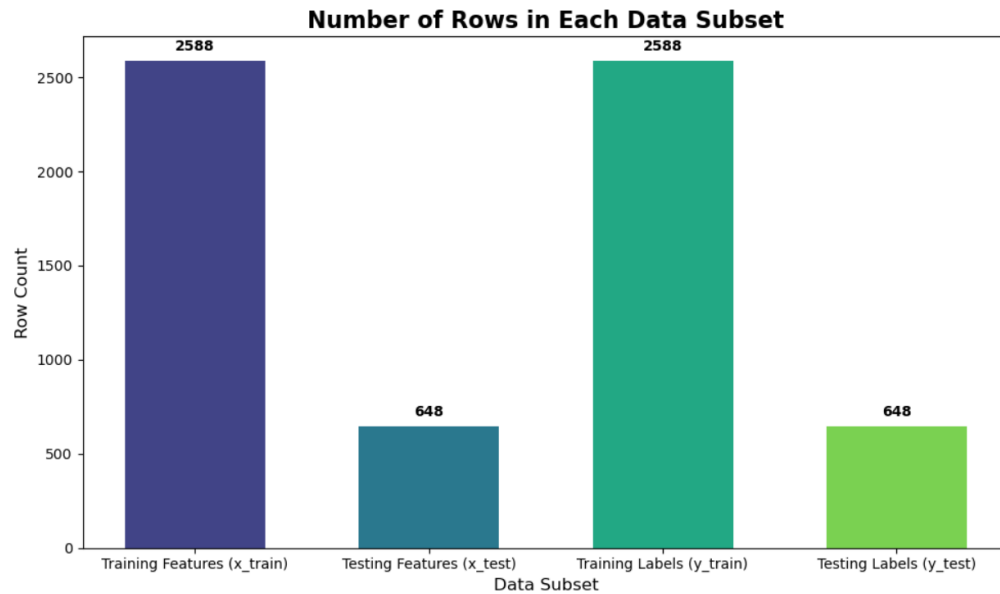
4. Splitting datasets into two sets. Document your rations for splitting and results.

- 4.1. We then split the data into our training and test data using the sklearn's built in `train_test_split()` function. This separated our datasets into 80% training data and 20% test or validation data. For the training dataset, we excluded the "Gender" column because it was irrelevant for our training model. We used stratified splitting to ensure that both the tests have the same proportion of classes. We chose our target variable as the "Recovered" column because we wanted to see if the patient had recovered or not through various treatments while afflicted with different types of lung disease. Finally, we shuffled the data before splitting to avoid any bias by mixing any ordering of data. The split accurately split our dataset into the models and displayed the results of what we wanted to see.

```
We have a total of 3236 rows of data.
80% of 3236 is 2588 which should be our testing data
Training would be 648

RESULTS OF SPLIT
Training set shape details are: x_train = (2588, 10), y_train= (2588,)
Testing set shape details are: x_test = (648, 10), y_train= (648,)

--- Training Set Statistics ---
      Age  Smoking Status  Lung Capacity  Hospital Visits  \
4763   50                1                1                9
2939   36                0                1                14
2106   65                1                1                9
1257   88                1                3               10
4598   73                1                4                3
```

4.2. We then calculated the mean and standard deviation for both the training and the test data to see if they matched. The results showed that they were extremely close, which meant that the split of the two datasets was representative and consistent.

```

The mean for the training data is Age                    54.235317
Smoking Status      0.509274
Lung Capacity       3.004637
Hospital Visits     7.494204
Disease Type_Bronchitis 0.207883
Disease Type_COPD    0.213679
Disease Type_Lung Cancer 0.186631
Disease Type_Pneumonia 0.194745
Treatment Type_Surgery 0.322643
Treatment Type_Therapy 0.334621
dtype: float64 and the standard deviation is Age        20.042337
Smoking Status      0.500011
Lung Capacity       1.421567
Hospital Visits     3.994532
Disease Type_Bronchitis 0.405871
Disease Type_COPD    0.409982
Disease Type_Lung Cancer 0.389690
Disease Type_Pneumonia 0.396081
Treatment Type_Surgery 0.467578
Treatment Type_Therapy 0.471949
dtype: float64
The mean for the testing data is Age                    54.206790
Smoking Status      0.526235
Lung Capacity       2.907407
Hospital Visits     7.533951
Disease Type_Bronchitis 0.205247
Disease Type_COPD    0.189815
Disease Type_Lung Cancer 0.194444
Disease Type_Pneumonia 0.197531
Treatment Type_Surgery 0.347222
Treatment Type_Therapy 0.319444
dtype: float64 and the standard deviation is Age        20.393512
Smoking Status      0.499697
Lung Capacity       1.447934
Hospital Visits     3.940683

```